Lab 7: Fuzz Testing for Input Validation Bugs

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Bike Share Rental Parser

SS1 - Buggy application run

```
SS4 - Hypothesis tests against fixed application

$ python -m pytest test_bike_share_rental_parser.py

============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\SE\SE-Experiential-Learning\L6_L7_53_bike_share_rental\LAB7
plugins: anyio-4.13.0, hypothesis-6.170.15
collected 3 items

test_bike_share_rental_parser.py ...                                     [100%]

============================== warnings summary ===============================
..\..\..\..\Python 3.12.10 (64-bit)\Lib\site-packages\anyio\pytest_plugin.py:14
  D:\Python 3.12.10 (64-bit)\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning:
_pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 3 passed, 1 warning in 1.83s =========================
```

SS2 - Hypothesis tests against buggy application

SS1 - Buggy application run

```
$ python bike_share_rental_parser.py

==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): Duration in minutes: 75

==== Running format_station_code ====
Enter a station code (e.g. a12): Formatted station code: A-12

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): Parsed ride log: ('electric',
45, True)
```

SS3 - Fixed application run

```
SS2 - Hypothesis tests against buggy application

$ python -m pytest test_bike_share_rental_parser.py

============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.0.0, pluggy-1.7.0
rootdir: C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy
plugins: anyio-4.13.0, hypothesis-6.170.15
collected 3 items

test_bike_share_rental_parser.py FFF                                     [100%]

================================== FAILURES ===================================
_____________________ test_parse_ride_duration_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py",
line 5, in test_parse_ride_duration_no_crash
  |     def test_parse_ride_duration_no_crash(s):
  |                ^^^
  | exceptiongroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 7, in
test_parse_ride_duration_no_crash
    |     parse_ride_duration(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 7, in parse_ride_duration
    |     return int(hours) * 60 + int(minutes)
    |                ^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 9, in
test_parse_ride_duration_no_crash
    |     assert False, f"parse_ride_duration crashed with: {exc}"
    | AssertionError: parse_ride_duration crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Falling test case: test_parse_ride_duration_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 7, in
test_parse_ride_duration_no_crash
    |     parse_ride_duration(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 6, in parse_ride_duration
    |     hours, minutes = text.strip().split(':')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 9, in
test_parse_ride_duration_no_crash
    |     assert False, f"parse_ride_duration crashed with: {exc}"
    | AssertionError: parse_ride_duration crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Falling test case: test_parse_ride_duration_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 7, in
test_parse_ride_duration_no_crash
    |     parse_ride_duration(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 6, in parse_ride_duration
    |     hours, minutes = text.strip().split(':')
    |                      ^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 9, in
test_parse_ride_duration_no_crash
    |     assert False, f"parse_ride_duration crashed with: {exc}"
    | AssertionError: parse_ride_duration crashed with: 'NoneType' object has no attribute 'strip'
    | assert False
    | Falling test case: test_parse_ride_duration_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_format_station_code_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py",
line 12, in test_format_station_code_no_crash
  |     def test_format_station_code_no_crash(s):
  |                ^^^
  | exceptiongroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 14, in
test_format_station_code_no_crash
    |     format_station_code(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 15, in format_station_code
    |     return code[0] + '-' + code[1:]
    |                      ~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 16, in
test_format_station_code_no_crash
    |     assert False, f"format_station_code crashed with: {exc}"
    | AssertionError: format_station_code crashed with: string index out of range
    | assert False
    | Falling test case: test_format_station_code_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |
C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py:15
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 14, in
test_format_station_code_no_crash
    |     format_station_code(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 14, in format_station_code
    |     code = code.strip().upper()
    |            ^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 16, in
test_format_station_code_no_crash
    |     assert False, f"format_station_code crashed with: {exc}"
    | AssertionError: format_station_code crashed with: 'NoneType' object has no attribute 'strip'
    | assert False
    | Falling test case: test_format_station_code_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |
C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py:15
    +------------------------------------
_____________________ test_parse_ride_log_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py",
line 19, in test_parse_ride_log_no_crash
  |     def test_parse_ride_log_no_crash(s):
  |                ^^^
  | exceptiongroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 21, in
test_parse_ride_log_no_crash
    |     parse_ride_log(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 23, in parse_ride_log
    |     return parts[0].strip(), int(parts[1]), parts[2].strip() == 'yes'
    |                                  ~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 21, in
test_parse_ride_log_no_crash
    |     assert False, f"parse_ride_log crashed with: {exc}"
    | AssertionError: parse_ride_log crashed with: list index out of range
    | assert False
    | Falling test case: test_parse_ride_log_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 21, in
test_parse_ride_log_no_crash
    |     parse_ride_log(s)
    |   File "c:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\bike_share_rental_parser.py",
line 22, in parse_ride_log
    |     parts = line.split(',')
    |             ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File
"C:\Users\harsh\AppData\Local\Temp\labz_pdf_uoezcwmi\buggy\test_bike_share_rental_parser.py", line 21, in
test_parse_ride_log_no_crash
    |     assert False, f"parse_ride_log crashed with: {exc}"
    | AssertionError: parse_ride_log crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Falling test case: test_parse_ride_log_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
============================== warnings summary ===============================
D:\Python 3.12.10 (64-bit)\Lib\site-packages\anyio\pytest_plugin.py:14
  D:\Python 3.12.10 (64-bit)\Lib\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning:
_pytest.python.CallSpec2 has been renamed to CallSpec.
    The CallSpec2 alias will be removed in pytest 10.
    Update imports to use CallSpec instead.
    See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================== short test summary info ===========================
FAILED test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash - ...
FAILED test_bike_share_rental_parser.py::test_format_station_code_no_crash - ...
FAILED test_bike_share_rental_parser.py::test_parse_ride_log_no_crash - Excep...
======================== 3 failed, 1 warning in 3.56s =========================
```

SS4 - Hypothesis tests against fixed application

```
SS3 - Fixed application run

$ python bike_share_rental_parser.py

==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): Duration in minutes: 75

==== Running format_station_code ====
Enter a station code (e.g. a12): Formatted station code: A-12

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): Parsed ride log: ('electric',
45, True)
```

Reflection Questions and Answers

1. What bugs were identified?

The original parser assumed every value had the expected type and format. Empty strings caused indexing errors, missing fields caused split/index errors, non-numeric minutes caused ValueError, and None values caused AttributeError when string methods were called.

2. How did Hypothesis help?

Hypothesis generated many arbitrary Unicode strings and None values automatically. This exposed minimal failing examples that ordinary valid-input tests would not cover, allowing the tests to verify the broader property that each public parser function does not crash on unexpected input.

3. How were the bugs fixed?

Each function now validates the input type and structure before processing it. Malformed duration values return 0, empty or non-string station codes return an empty string, and malformed ride logs return the neutral tuple ("", 0, False). Valid documented inputs retain their original behavior.

Validation result: all three Hypothesis tests pass on the fixed application.